

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase 3, Randomized, Open-label, Active-Comparator-Controlled Clinical Study to Evaluate the Safety and Efficacy of Bomedemstat (MK-3543/IMG-7289) Versus Best Available Therapy (BAT) in Participants With Essential Thrombocythemia Who Have an Inadequate Response to or Are Intolerant of Hydroxyurea
Short Title:	A Study of Bomedemstat (IMG-7289/MK-3543) Compared to Best Available Therapy (BAT) in Participants With Essential Thrombocythemia and an Inadequate Response or Intolerance of Hydroxyurea (MK-3543-006)
Protocol Number:	1015841870201971649
NCT Number:	NCT06079879
Sponsor Name:	Merck
Investigational Product:	busulfan
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Durable Clinicohematologic Response (DCHR) Rate: DCHR rate is the percentage of participants with DCHR, defined as a confirmed reduction of platelet count to $\leq 400 \times 10^9/L$, absence of white blood cell (WBC) count elevation to $> 10 \times 10^9/L$ locally assessed to be due to ET, and the absence of any thrombotic or major hemorrhagic events or disease progression to myelofibrosis (MF), myelodysplastic syndrome (MDS) or acute myeloid leukemia (AML) by Week 52.
Secondary Obj.:	1. Change From Baseline in Myelofibrosis Symptom Assessment Form (MFSAF) v4.0 Individual Fatigue Symptom Item Score 2. Change From Baseline in Patient-reported Outcomes Measurement Information System (PROMIS) Fatigue SF-7a Total Fatigue Score 3. Change From Baseline in Total Symptom Score as Measured on the MFSAF v4.0

2.2 Background & Rationale

This is a study evaluating the safety and efficacy of bomedemstat (MK-3543) compared with the best available therapy (BAT) in participants with essential thrombocythemia (ET) who have an inadequate response to or are intolerant of hydroxyurea. The primary study hypothesis is that bomedemstat is superior to the best available therapy with respect to durable clinicohematologic response (DCHR).

3. Study Design & Methodology

Design Framework:	Type: INTERVENTIONAL, Phase: PHASE3, Status: RECRUITING
Target N:	340

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- * Has a diagnosis of ET per WHO 2016 diagnostic criteria for myeloproliferative neoplasms (confirmed by a central pathologist)
- * Has a centrally assessed bone marrow fibrosis score of Grade 0 or Grade 1, as per a modified version of the European Consensus Criteria for Grading Myelofibrosis
- * Has a history of inadequate response to or intolerance of hydroxyurea based on modified European LeukemiaNet (ELN) criteria for hydroxyurea resistance or intolerance
- * Has an inadequate or loss of response to their most recent prior ET therapy, requiring a change of cytoreductive therapy
- * Has a platelet count $> 450 \times 10^9/L$ (450k /?L) assessed up to 72 hours before first dose of study intervention
- * Has an absolute neutrophil count (ANC) $\geq 0.75 \times 10^9/L$ assessed up to 72 hours before first dose of study intervention
- * Participants may have received up to 3 prior ET-directed cytoreductive agents including hydroxyurea

4.2 Exclusion Criteria

- * Known immediate or delayed hypersensitivity reaction or idiosyncrasy to drugs chemically related to bomedemstat or lysine demethylase or monoamine oxidase inhibitor (LSDi or MAOi) or the chosen best available therapy (including anagrelide, interferon alfa/pegylated interferon, ruxolitinib, or busulfan) that contraindicates participation
- * History of any illness/impairment of GI function that might interfere with drug absorption (eg, chronic diarrhea or history of gastric bypass surgical procedure), confound the study results or pose an additional risk to the individual by participation in the study
- * Evidence at the time of Screening of increased risk of bleeding
- * History of a malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years. Note: The time requirement does not apply to participants who underwent successful definitive resection of basal cell carcinoma of the skin, squamous cell carcinoma of the skin, or carcinoma in situ, excluding carcinoma in situ of the bladder
- * Human immunodeficiency virus (HIV)-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease

11. Study Identifiers & Governance

Primary ID:	1015841870201971649
Registry Number:	NCT06079879
Sponsor:	Merck
Study Title:	A Study of Bomedemstat (IMG-7289/MK-3543) Compared to Best Available Therapy (BAT) in Participants With Essential Thrombocythemia and an Inadequate Response or Intolerance of Hydroxyurea (MK-3543-006)

15. Sign-off & Approval

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Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____